

修士論文

Microlocal Morse Theory and Truncated  
Microsupport  
(超局所モース理論と打ち切り超局所台)

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## 1 Introduction

In this paper, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

### 1.1 Organization

This paper is organized as follows. In Section 5, we prove a truncated version of the noncharacteristic deformation lemma.

## 2 Notations and Reviews

We consider a commutative unital ring  $\mathbf{k}$  of finite global dimension (e.g.  $\mathbf{Z}$  or a field). Let  $X$  be a topological space. We denote by  $D^b(\mathbf{k}_X)$  the bounded derived category of sheaves of  $\mathbf{k}$ -modules on  $X$ .

Let us denote  $Z$  a locally closed set of  $X$  and  $i: Z \hookrightarrow X$  an inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\cdot) := \mathrm{R}i_*i^!, \quad (\cdot)_Z := \mathrm{R}i_!i^{-1}.$$

If there is no risk, we will write  $\mathbf{k}_Z$  rather than  $\mathbf{k}_{X,Z}$ .

### Geometrical Notations

Let  $M$  be a real manifold of class  $C^\infty$ . We denote by  $\pi: T^*M \rightarrow M$  the cotangent bundle of  $M$ . If  $f: M \rightarrow N$  is a morphism of manifolds, then we have the natural diagram.

$$\begin{array}{ccccc} TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\ & \searrow \tau_M & \downarrow & \square & \downarrow \tau_N \\ & & M & \xrightarrow{f} & N, \end{array} \quad \begin{array}{ccccc} T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\ & \searrow \pi_M & \downarrow & \square & \downarrow \pi_N \\ & & M & \xrightarrow{f} & N. \end{array}$$

In this diagram,  $f_\tau$  and  $f_\pi$  denote the natural projections.  $f'$  is defined by  $(x; v) \mapsto (x; df_x v)$ , and  $f_d$  is induced by the transposition of  $f'$ .

### 3 Truncated Microsupport

**Definition 3.1.** Let  $F \in \mathrm{D}^b(\mathbf{k}_X)$ . The closed conic subset  $\mathrm{SS}_k(F)$  of  $T^*X$  is defined by:  $p \notin \mathrm{SS}_k(F)$  if and only if there exists an open conic neighborhood  $U$  of  $p$  such that for any  $x \in \pi(U)$  and for any real-valued  $C^1$ -function  $\varphi$  defined on a neighborhood of  $x$  such that  $\varphi(x) = 0$ ,  $d\varphi(x) \in U$ , one has

$$(3.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

### Functorial Properties

**Proposition 3.2.** Let  $f: X \rightarrow Y$  be a morphism of real analytic manifolds and let  $F \in \mathrm{D}^b(\mathbf{k}_X)$  such that  $f$  is proper on  $\mathrm{supp}(F)$ . Then for any  $k \in \mathbf{Z}$ ,

$$(3.2) \quad \mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)).$$

The equality holds in case  $f$  is a closed embedding.

### 4 Lemmas from [KS90]

Consider a projective system of abelian groups  $(M_n, \rho_{n,p})_n$  indexed by  $\mathbf{N}$ . One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any  $n \in \mathbf{N}$ , the decreasing sequence  $(\rho_{n,p}(M_p))_p$  of subgroups of  $M_n$  is stationary.

For a projective system of abelian groups  $(M_n)_n$ , we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(4.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(4.2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(4.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

**Proposition 4.1** ([KS90, Proposition 1.12.4]). Assume that for each  $j \in \mathbf{Z}$ , the system  $(M_n^{j-1})_n$  satisfies the ML condition. Then

- (a) for each  $k \in \mathbf{Z}$ , the map  $\Phi_k$  is surjective,
- (b) if moreover, for a given  $i$  the system  $(H^{i-1}(E_n^\bullet))_n$  satisfies the ML condition, then  $\Phi_i$  is bijective.

The following is called Kashiwara's lemma.

**Proposition 4.2** ([KS90, Proposition 1.12.6]). Let  $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$  be a projective system of sets indexed by  $\mathbf{R}$ . Assume that for each  $s \in \mathbf{R}$ , the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps  $\rho_{s_0, s_1}$  ( $s_0 \leq s_1$ ) are injective (resp. surjective).

## ML procedure for sheaves

**Proposition 4.3** ([KS90, Proposition 2.7.1]). Let  $X$  be a topological space and let  $F \in \mathbf{D}^+(\mathbf{Z}_X)$ . Let  $(U_n)_{n \in \mathbf{N}}$  be an increasing sequence of open subsets of  $X$ ,  $(Z_n)_{n \in \mathbf{N}}$  a decreasing sequence of closed subsets of  $X$ . Set  $U := \bigcup_n U_n$ ,  $Z := \bigcap_n Z_n$ .

- (i) For any  $j$ , the natural map  $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$  is surjective.
- (ii) Assume that for a given  $j$ , the projective system  $(H_{Z_n}^{j-1}(U_n; F))_n$  satisfies the ML condition. Then  $\varphi_j$  is bijective.
- (iii) Let now  $(X_n)_{n \in \mathbf{N}}$  be an increasing family of subsets of  $X$  satisfying  $X = \bigcup_n X_n$  and  $X_n \subset \text{Int}(X_{n+1})$  for all  $n$ . Assume that for a given  $j$ , the projective system  $(H^{j-1}(X_n; F))_n$  satisfies the ML condition. Then the natural map  $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$  is bijective.

## 5 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

**Proposition 5.1.** Let  $X$  be a locally compact (i.e., by definition, Hausdorff) space countable at infinity (i.e., countable union of compact subspaces). Let  $F \in \mathbf{D}^b(\mathbf{Z}_X)$ , and  $(U_t)_{t \in \mathbf{R}}$  be a family of open subsets of  $X$ . Let  $k \in \mathbf{Z}$ . If  $s$  is a real number, set  $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$ . We assume the following conditions.

- (i)  $U_t = \bigcup_{s < t} U_s$ , for all  $t \in \mathbf{R}$ .
- (ii) For all pairs  $(s, t)$  with  $s \leq t$ , the set  $\overline{U_t - U_s} \cap \text{supp } F$  is compact.
- (iii) For all pairs  $(s, t)$  with  $s \leq t$ , and all  $x \in Z_s - U_t$ , we have

$$(H_{X-U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For any  $s \in \mathbf{R}$ ,  $Z_s \subset U_t$ .

Then, for all  $t \in \mathbf{R}$ , we have the isomorphisms

$$H^j \left( \bigcup_{s \in \mathbf{R}} U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left( \bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_t; F).$$

**Remark 5.2.** Let us give an example that satisfies the hypotheses (i)–(iii) but not (iv). Let us define a locally closed subset  $Z$  of  $\mathbf{R}^2$  by

$$Z := \{(x, y) \in \mathbf{R}^2 ; (x+2)^2 + (y-1)^2 = 1\} \setminus \{(-2, 0)\}.$$

Consider a real valued function  $\varphi_s$  for a real number  $s$ :

$$(5.1) \quad \varphi_s(x) := \begin{cases} 0 & |x| \geq 2, \\ \max(0, s(2 - |x|)) & |x| \leq 2. \end{cases}$$

We define  $(U_t)_{t \in \mathbf{R}}$ , a family of open sets of  $\mathbf{R}^2$  by

$$U_t := \{(x, y) \in \mathbf{R}^2 ; y < \varphi_s(x)\}.$$

Then  $F := \mathbf{k}_Z$  and  $(U_t)_t$  satisfy the hypotheses for  $k = 0$ . Therefore, the natural morphism

$$H^0(U_1; \mathbf{k}_Z) = \mathbf{k} \rightarrow H^0(U_0; \mathbf{k}_Z) = 0$$

is not injective and the consequence of Proposition 5.1 does not hold.

**Proof of Proposition 5.1.** Consider the following conditions for  $j \in \mathbf{Z}$ ,  $s$ , and  $t$ :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s<t} H^j(U_s; F). \end{aligned}$$

Let us prove  $(a)_s^j$  for  $s \in \mathbf{R}$  and  $j < k$ . We can assume that  $\overline{U_t - U_s}$  is compact for any  $s \leq t$  by replacing  $X$  by  $\text{supp } F$ . Consider the distinguished triangle

$$\text{R}\Gamma_{X-U_t}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Applying to this triangle the functor  $(\cdot)_{\overline{U_t}}$ , we have

$$(\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}} \rightarrow F_{\overline{U_t}} \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor  $\text{R}\Gamma(X; \cdot)$ , we have

$$(5.2) \quad \text{R}\Gamma(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) \rightarrow \text{R}\Gamma(\overline{U_t}; F) \rightarrow \text{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

By hypotheses (iii)–(??), it follows that

$$\left( H_{X-U_t}^j(F) \right)_{\partial U_t} = 0$$

for any  $j \leq k$  and then

$$H^j(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) = 0 \quad \text{for any } j \leq k.$$

\*1 Thus for any  $t$ , we have

$$(5.3) \quad H^j(\overline{U_t}; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(5.4) \quad H^k(\overline{U_t}; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (5.3),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) & \cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ & = H^j(\overline{U_s}; F) \\ & \xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any  $j < k$ , and  $(a)_s^j$  follows. Similarly, we have by (iv) and (5.4),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) & \cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ & = H^k(\overline{U_s}; F) \\ & \hookrightarrow H^k(U_s; F) \end{aligned}$$

and  $(a')_s^k$  follows. □

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\*1 この辺の議論はきちんと詰めないといけない。

## 6 Relative Microlocal Morse Theory via Truncated Microsupport

Let  $f: Y \rightarrow X$  be a morphism of manifolds, and  $\varphi$  a real valued function of class  $C^1$ . For a real number  $t \in \mathbf{R}$ , we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}((-\infty, t)), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}((-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

**Proposition 6.1.** Let  $G \in D^b(Y)$ , and assume  $f_t: \text{supp}(G) \cap \bar{Y}_t \rightarrow X$  is proper for all  $t \in \mathbf{R}$ . Let  $k$  be an integer and  $t_o$  a real number.

(i) Assume that, for any  $y \in Y - \tilde{Y}_{t_o}$ , we have

$$d\varphi(y) \notin \left( \text{SS}_k(G) + f_d \left( Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(a)} \quad & R^j f_* G \cong R^j f_{t*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(a}_k\text{)} \quad & R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(a')} \quad & R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(a}'_k\text{)} \quad & R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(b)} \quad & \text{SS}_k(Rf_* G) \subset f_\pi \left( f_d^{-1} \left( \text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any  $y \in Y - \tilde{Y}_{t_o}$ , we have

$$-d\varphi(y) \notin \left( \text{SS}_k(G) + f_d \left( Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(c)} \quad & R^j f_! G \cong R^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(c}_k\text{)} \quad & R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(c')} \quad & R^j \tilde{f}_! G \cong R^j \tilde{f}_{t!} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(c}'_k\text{)} \quad & R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t!} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(d)} \quad & \text{SS}_k(Rf_! G) \subset f_\pi \left( f_d^{-1} \left( \text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & \subset f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$



**Proof.** Let  $\tilde{f}$  be the morphism  $(f, \varphi): Y \rightarrow X \times \mathbf{R}$ , and let  $q$  and  $\tilde{\varphi}$  be the first and second projections defined on  $X \times \mathbf{R}$  respectively. Then  $\tilde{f}$  is proper on  $\text{supp}(G)$  and  $f = q \circ \tilde{f}$ . This can be proved as follows. If  $K$  is a compact set of  $X \times \mathbf{R}$ ,  $\tilde{f}^{-1}(K)$  is compact.

Let  $i_t$  be the open embedding  $X \times (-\infty, t) \hookrightarrow X \times \mathbf{R}$  and  $\tilde{i}_t$  the closed embedding  $X \times (-\infty, t] \hookrightarrow X \times \mathbf{R}$ . Set  $q_t = q \circ i_t$  and  $\tilde{q}_t = q \circ \tilde{i}_t$ . These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ \uparrow j_t & & \uparrow i_t \nearrow q_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times (-\infty, t) \end{array} \quad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X. \\ \uparrow \tilde{j}_t & & \uparrow \tilde{i}_t \nearrow \tilde{q}_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times (-\infty, t] \end{array}$$

Then, setting  $\tilde{G} := Rf_*G$ , we have

$$\begin{aligned} Rf_*G &\cong Rq_*\tilde{G}, & R\tilde{f}_*G &\cong R\tilde{q}_*\tilde{G}, \\ Rf_{t*}j_t^{-1}G &\cong Rq_{t*}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R\tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ Rf_{t!}j_t^{-1}G &\cong Rq_{t!}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^!G &\cong R\tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Thus taking  $j$ -th cohomology, we have

$$\begin{aligned} R^j f_*G &\cong R^j q_*\tilde{G}, & R^j \tilde{f}_*G &\cong R^j \tilde{q}_*\tilde{G}, \\ R^j f_{t*}j_t^{-1}G &\cong R^j q_{t*}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ R^j f_{t!}j_t^{-1}G &\cong R^j q_{t!}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^!G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for  $\tilde{G}$ ,  $q$ ,  $i_t$ ,  $\tilde{\varphi}$  instead of  $G$ ,  $f$ ,  $j_t$ ,  $\varphi$ .

(i) It follows from the hypothesis that

$$(6.1) \quad \text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any  $(y; \eta) \in \text{SS}_k(G)$  and  $(f(x); \xi) \in Y \times_X T^*X$ , we have

$$(6.2) \quad d\varphi(y) \neq \eta + f^*\xi.$$

If  $(x, t; \xi, \tau)$  is a point of  $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$ , then, by the formula (3.2), there exists  $(y; \eta) \in \text{SS}_k(G)$  such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

For this  $\eta$ , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If  $\tau > 0$ , we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (6.2) because  $\text{SS}_k(G)$  is conic.

By (6.1), Let  $U$  be an open subset of  $X$  contained in a local chart, and convex in such a chart. By Proposition 5.1, we have isomorphisms for all  $t > t_o$  and  $j < k$

$$(6.3) \quad H^j(U \times \mathbf{R}; \tilde{G}) \xrightarrow{\sim} H^j(U \times (-\infty, t); \tilde{G})$$

and have an injection for all  $t > t_0$

$$(6.4) \quad H^k(U \times \mathbf{R}; \tilde{G}) \hookrightarrow H^k(U \times (-\infty, t); \tilde{G}).$$

This can be showed as follows.

This proves the result. □

**Remark 6.2.**  $(U \times I_t)$  is not a gentle family.

## 参考文献

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